Creating a Harmonized Dataset by Merging Multiple Perfume Databases

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Abstract. This project focuses on the creation of a unified dataset for perfume analytics by integrating data from multiple sources. The primary objective was to ensure that each perfume is uniquely represented by a single, conflict-free record. To achieve this, data consolidation techniques were applied to resolve inconsistencies and standardize attributes, enhancing the dataset's quality and usability. The integration process facilitated the elimination of redundancies and provided a holistic view of the data, enabling more accurate analysis and decision-making. This unified dataset serves as a reliable foundation for downstream applications, such as trend analysis, and product comparisons, demonstrating the importance of data harmonization in complex domains.

Keywords: Data Integration · Data Consistency · Unique Records

1 Introduction

Integrating data from multiple sources is essential to obtain a unified view as it allows for a comprehensive and accurate representation of information. Different sources often contain fragmented or complementary data that, when combined, provide a fuller picture. Without integration, inconsistencies, gaps, and redundancies can arise, limiting the reliability and usability of the data. A unified dataset ensures consistency, resolves conflicts, and enhances data quality, enabling more effective analysis and decision-making. This approach is especially critical in domains like perfume analytics, where diverse attributes and descriptors from various origins must be harmonized to capture a holistic understanding.

The objective of this project is to develop a comprehensive merged dataset where each perfume is uniquely represented by a single record, ensuring consistency and eliminating data conflicts. This involved consolidating information from multiple sources, resolving discrepancies, and standardizing data fields to achieve uniformity. By addressing redundancies and conflicting entries, the final dataset provides a reliable foundation from analysis, facilitating a clear and accurate understanding of the represented perfumes.

2 Phase I: Data Collection

The data for this analysis stems from three datasets, each contributing in perspectives and details about perfumes. A description of the datasets can be found in table 1. The list of attributes of each dataset can be found in table 2.

Table 1. Datasets

Dataset	# of entities	# of attributes	format	source
perfume database	37926	8	xlsx	GitHub[1]
pefume	26319	11	xls	Hugging Face[2]
parfumo datos	59325	12	csv	Kaggle[3]

Table 2. List of attributes

Dataset	List of attributes
perfume database	brand, perfume, image, launch year, main accords, notes, longevity, sillage
pefume	brand, name perfume, family, subfamily, fragances, ingredients, origin, gender, years, description, image name
parfumo datos	Name, Brand, Release Year, Concentration, Rating Value, Rating Count, Main Accords, Top Notes, Middle Notes, Base Notes, Perfumers, URL

Across the datasets, the attributes brand + perfume and brand + name perfume and Brand + Name are the identifying attributes. The name value alone is not sufficient because two brands may publish a perfume with the same name and the brand attribute alone is not sufficient because a brand produces multiple perfumes. Those attributes are also never missing.

The following attributes are discarded because they cannot be matched to any attribute in other datasets: image, longevity, and sillage in perfume database, origin, gender, description, and image name in pefume, and rating value, rating count, perfumers, and URL in parfumo datos.

In the dataset perfume database, the value for the release year is missing in 30% of all entities, and the values for main accords and notes are missing in 3% of all entities each. There is a high correlation between whether a value is missing for the main accords and for the notes.

In the dataset pefume, all relevant attributes are never missing.

[1] https://github.com/rdemarqui/perfume_recommender/blob/main/database/perfume_database.xlsx
[2] https://huggingface.co/datasets/doevent/perfume/blob/main/pefume.xls
[3] https://www.kaggle.com/datasets/olgagmiufana1/parfumo-fragrance-dataset

In the dataset parfumo datos, the value for the release year is missing in 34% of all entities, the value for main accords is missing in 46% of all entities, and the values for top notes, middle notes, and base notes are missing in 47% of all entities. There is no clear correlation between whether a value is missing between the main accords and the notes, but the entities with missing values in top notes, middle notes, and base notes are quasi the same.

3 Phase I: Data Translation

Previous to the conversion of the datasets into an integrated schema using MapForce, we applied some preprocessing in Python. To do so, we read the files into a pandas dataframe, apply the preprocessing steps and print the results back to files, respectively. Preprocessing includes some basic text normalization steps like removing special characters, removing stop words, and lower casing. Additionally, we discovered that in parfumo datos, the name attributes contains the name of the brand, so we remove the brand out of the name value, and the brand attribute sometimes contains duplicates in different languages, so we remove these duplicates. For all other attributes which are not discarded, we perform lower casing.

To create a unified schema across all datasets, we use MapForce in a top-down approach. We created a target schema .xml file with two sample records and then mapped the attributes of the dataset to the target schema. The target schema contains the attributes brand, name, year, concentration, fragrance, main accords, and notes where main accords is a list attribute and notes contains three sub-attributes top notes, middle notes, and base notes, being a list attribute each.

Correspondences are as follows: brand, name, year, concentration, and fragrance are always 1:1 correspondences. We know that concentration and fragrance are only present in one of the three datasets each, but their values might help to perform identity resolution later which is definitely not the case for the discarded attributes.

For the dataset perfume database, the attribute main accords is tokenized and mapped as a list attribute in the target schema. The attribute notes contains in most cases as values a dictionary-like structure with "top", "middle", and "base" as keys and their respective list as values. Thus, we split the value by these keys, tokenize the value and map it as a list attribute into the target schema. In some cases, however, it just contains a list of notes, so we map it to the super-attribute note and not to a top, middle, or base note. For the dataset pefume, the attributes family and subfamily are mapped to main accords in the target schema because they contain similar values (i.e., it is instance-based mapping). If family and subfamily contain the same value, only one of them is mapped to main accords. If not, the values are concatenated and mapped to main accords. The attribute ingredients does not contain any information about top, middle, and base notes, and thus is mapped to notes. Again, this is due to the values because of instance-based mapping. For the dataset parfumo datos, the attributes main accords, top notes, middle notes, and base notes are tokenized and then mapped as the list attribute in the target schema.

At the end of this schema mapping, we now have three .xml files each with a single unified schema. In addition to the correspondences, all entities are assigned a globally unique ID.

4 Phase II: Identity Resolution

Identity resolution in our project focused on reconciling and linking records of perfume entities across three datasets: Pefume, Perfume Database, and Parfumo Datos. This phase aimed to identify similar entities, consolidate data, and create a unified schema to facilitate integrated analysis. We used the Winte.R framework for identity resolution, resulting in an integrated XML schema.

4.1 Identifying Similar Entities

The primary task was to match perfume records across the datasets. Each dataset contained overlapping but distinct information about perfumes, such as brand names, product names, and other attributes. Records were linked by comparing attributes using similarity measures and optimizing the process through blocking techniques to reduce the number of unnecessary comparisons.

4.2 Gold Standard

Our gold standard dataset consists of a total of 1,200 entries, distributed across 500 comparisons between Parfumo Datos and the Perfume Database, 500 comparisons between Parfumo Datos and Pefume, and 200 comparisons between the Perfume Database and Pefume. The distribution between true matches and non-matches in the gold standard is balanced, with 600 matches and 600 non-matches.

We created the gold standard dataset by manually comparing records from the raw data files before performing identity resolution. Therefore, we believe there is no selection bias. Furthermore, the gold standard includes a fair number of edge cases on both the match and non-match sides.

4.3 Similarity Measures

We tested various combinations of similarity measures to evaluate the effectiveness of matching algorithms:

- **Jaccard similarity:** Compared sets of tokenized attributes (e.g., brand and name).
- **Levenshtein Distance:** Assessed the edit distance between textual attributes to capture typographical variations.
- **Hybrid Measures (Jaccard + Levenshtein/2):** Combined measures to enhance accuracy for complex fields.

These measures were applied to different combinations of attributes, such as brand names, perfume names.

4.4 Matching Rules and Experimental Setup

Once blocking grouped records into manageable sets, we applied a linear combination matching rule with weightage of 0.5 and then applied a set of comparators to identify and reconcile similar entities. These comparators combined different similarity measures and weighting schemes to optimize the accuracy of matches.

- **Comparator 1:**
  - *Attributes Used:* Brand Name (weighted 0.4) and Perfume Name (weighted 0.6)
  - *Similarity Measure:* Jaccard Similarity
  - *Description:* This rule evaluated the overlap of tokenized sets for both brand and perfume names. The perfume name was given higher importance due to its higher likelihood of being distinctive compared to the brand name.
- **Comparator 2:**
  - *Attributes Used:* Brand Name (weighted 0.4) and Perfume Name (weighted 0.6)
  - *Similarity Measure:* Levenshtein Distance
  - *Description:* This rule replaced Jaccard Similarity with Levenshtein Distance to account for minor typographical errors in brand and perfume names. By focusing on edit distance, it aimed to capture subtle variations that the token-based Jaccard method might miss.
- **Comparator 3:**
  - *Attributes Used:* Concatenated String of Brand Name + Perfume Name (weighted 1)
  - *Similarity Measure:* Hybrid (Jaccard Similarity + Levenshtein Distance / 2)
  - *Description:* This rule concatenated the brand and perfume names into a single string for comparison. A hybrid similarity measure combining Jaccard and Levenshtein (with equal weighting) was applied to this concatenated string. This approach aimed to capture both token overlap and edit distance in a holistic manner, ensuring robust matching for complex fields.

4.5 Result Analysis

The three comparators were tested across two dataset combinations:

1. Pefume and Parfumo datos
2. Perfume database and Parfumo datos

The performance of each comparator was evaluated using the standard metrics of precision, recall, and F1 score. The results for both dataset combinations are summarized1.

Key insights about the results

- **Blocking Efficiency:** Substring blocking based on brand names proved to be a practical solution after the challenges with year-based blocking. It balanced computational efficiency and accuracy while mitigating the impact of variability in brand name representation.
- **Performance of Comparator:**
  - Rule 1 (Jaccard Similarity) achieved high recall, making it suitable for scenarios where missing potential matches is critical.
  - Rule 2 (Levenshtein Distance) demonstrated strong performance in capturing subtle textual variations but showed slightly lower precision compared to Rule 1.
  - Rule 3 (Hybrid Similarity) provided a balanced performance, achieving a good trade-off between precision and recall. It was particularly effective for complex text fields where both token overlap and edit distance mattered.
- **Best-Performing Comparator:** While Rule 1 showed the highest recall, Rule 3's balanced F1 score made it the most reliable approach overall, particularly for datasets with diverse representations of brand and perfume names.

Comparator	Precision	Recall	F1 Score	Dataset Combination
Rule 1: Jaccard (Brand + Name, weighted 0.4/0.6)	0.7176	0.9261	0.8086	Pefume and Parfumo datos
Rule 2: Levenshtein (Brand + Name, weighted 0.4/0.6)	0.6727	0.9212	0.7775	Pefume and Parfumo datos
Rule 3: Hybrid (Brand+Name concatenated)	0.7227	0.8473	0.78	Pefume and Parfumo datos
Rule 1: Jaccard (Brand + Name, weighted 0.4/0.6)	0.7048	0.796	0.7477	Perfume database and Parfumo datos
Rule 2: Levenshtein (Brand + Name, weighted 0.4/0.6)	0.6763	0.8109	0.7376	Perfume database and Parfumo datos
Rule 3: Hybrid (Brand+Name concatenated)	0.7104	0.8296	0.7653	Perfume database and Parfumo datos

Fig. 1. Result Analysis

4.6 Blocking Strategy

The identity resolution process relied on blocking as a key optimization technique. Blocking reduces the comparison space by grouping records with similar attributes, ensuring only relevant candidates are matched.

We used year as the blocking attribute. The idea was to group perfumes based on their production or release year, as this field often distinguishes different entries. However, this approach encountered challenges because not all XML objects in the datasets included the year attribute. The absence of this crucial information for some records led to missed potential matches and gaps in the resolution process.

To address this issue, we pivoted to using brand names as the blocking attribute. While this approach resolved the issue of missing year information, it introduced its own challenges due to variability in how brand names were recorded across datasets (e.g., abbreviations, typographical errors, or additional descriptors). To further refine the process, we employed substring blocking, considering the first two characters of the brand name (substring(brand, 0, 2)). This adjustment successfully reduced the number of false negatives while maintaining efficiency.

For this project, due to time constraint, we haven't used no blocking strategy.

5 Phase III: Data Fusion

To establish the relationships between the data, we use the Jaccard metric, which has proved to offer the best results. We defined a gold standard of 500 records to assess the quality of the merge, applied to the correspondences between *perfume_ database.xml* and *perfumo_ datos.xml*, as well as between *perfume.xml* and *perfumo_ datos.xml*. For each attribute, specific merge rules are applied:

- **name**, **brand**, **years**: the shortest string is retained.
- **main accord**, **fragrance**, **note**: values from all three datasets are combined.

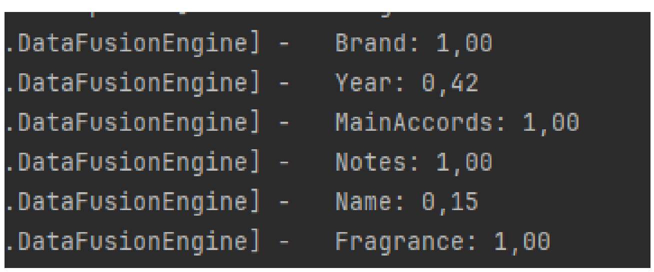

Fig. 2. Result from the data fusion engine

The merge evaluation focuses on the attributes name, brand, years, and if these match, the entire data set, including fragrance informations, is merged without further verification. The *gold_standard.xml* file, containing 15 examples refering to the three datasets at the same time, is used as a reference to test the consistency of the merge.
The attributes present in our gold_standard.xml are as follows:

- brand
- name
- year
- fragrance
- main accords
- notes

Concerning the results, according to the file produced by the merge *recordGroupConsistenciesV2.csv*, the consistency of the record groups is around 0.6 on average. The best results can reach an accuracy of 0.9 to 1. However, a limitation arises due to the reliance solely on the attributes **name**, **year**, and **brand** for similarity. This can lead to incorrect merges where two different perfumes are fused into a single record. This issue highlights the need for additional validation steps or more granular attributes to enhance differentiation3.


```xml
<perfume>
  <id>parfumo_datos_17656+parfumo_datos_41943+parfumo_datos_41944+parfumo_datos_56662+pefume_13458+pefume_18247+pefume_5799</id>
  <brand provenance="pefume_18247">berdoues</brand>
  <name provenance="pefume_18247">monoi tiare</name>
  <year provenance="parfumo_datos_56662"/>
  <fragrance provenance="pefume_18247">Floral Amber Fresher White Flower</fragrance>
  <main_accords provenance="parfumo_datos_41944">
    <accord>[, floral, , sweet, , creamy, , synthetic, , fresh, ]</accord>
  </main_accords>
  <notes provenance="pefume_5799">
    <note>[, bergamot, , lavender, , musk, , amber, , jasmine, , orange blossom, , rhubarb, , woody notes, , blackcurrant bud, ]</note>
  </notes>
</perfume>
```


Fig. 3. Example data from the fuse dataset

6 Conclusion

In this project, we successfully created a relatively harmonized dataset from three perfume databases using identity resolution (linear combination) and merging techniques. The main objective was to merge these different sources while ensuring that each perfume was represented uniquely, without conflict or duplication.

Through three main phases: data collection, identity resolution, and data merging. We developed different approaches to handle inconsistencies, missing values, and format differences. The identity resolution phase was particularly important as it enabled reliable linking of similar entities using comparison rules and a "gold standard" of 1200 manually verified entities (500 + 500 + 200).

The results showed that using different similarity measures, such as Jaccard similarity, Levenshtein distance, and a hybrid comparison approach combining Jaccard and Levenshtein, allowed for a data merge with an average consistency rate of approximately 0.75 for the best metric (Jaccard similarity). However, limitations remain, particularly due to the reliance on the **name**, **year** and **brand** attributes for merging, which can lead to errors in cases where two similar but distinct perfumes have limited information. One exploration avenue was considered regarding the main notes or accords, but due to time constraints, we were unable to implement this.

References

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